AS + A2 · Units 3 & 6 · WPH13 / WPH16 | Practical Skills

Practical Skills & Uncertainties

Units 3 and 6 are written papers about experiments. There's no practical exam, but you get questions on planning an experiment, taking readings, spotting what an inexperienced student did wrong, and processing real results. The skills on this page are the same for every topic, so learning them once picks up marks in every unit.

Unit 3 (WPH13)

Practical skills from Units 1 and 2: planning, implementing and processing results. You need percentage uncertainties from a **single** reading or from repeats, but you are **not** expected to combine (compound) uncertainties.

Unit 6 (WPH16)

The same skills applied to Units 4 and 5, plus **compounding percentage uncertainties**, **logarithmic graphs**, and the words precision, accuracy and sensitivity.

The language of measurement

Edexcel uses these words in a precise, scientific way (from the ASE's *The Language of Measurement*). Exam answers that use them loosely, like "the results were accurate because I repeated them", lose marks. Repeating improves **precision**, not accuracy.

Term	What it means
Validity	A measurement is valid if it measures what it is supposed to be measuring. This depends on both the method and the instruments.
True value	The value you'd get from an ideal measurement. Apart from fundamental constants, it is considered unknowable.
Accuracy	How close a result is judged to be to the true value. It can't be given a number, and it is affected by both random and systematic errors.
Precision	How closely repeated measurements agree with each other (how tightly they cluster). Only random effects affect it.
Repeatability	Precision when the same person uses the same method and equipment over a short time.
Reproducibility	Precision when different people use different apparatus or methods. A harder test than repeatability.
Uncertainty	The interval within which the true value can be considered to lie, e.g. 20 °C ± 2 °C.

Error	The difference between the measurement and the true value. It is *not* a mistake.
Resolution	The smallest measuring interval of an instrument, and the source of uncertainty in a single reading.

Random and systematic errors

RANDOM ERROR

Readings scatter **unpredictably** either side of the true value: reaction time with a stopwatch, a fluctuating meter, judging when a pendulum passes a point. They reduce **precision**.

Reduce it by repeating and averaging, timing many oscillations rather than one, or using a light gate or data logger instead of a human.

SYSTEMATIC ERROR

Every reading is wrong by the **same amount or the same proportion** in the same direction: a **zero error**, a stretched tape measure, always reading a scale from above (parallax), heat lost to the surroundings. They reduce **accuracy**, and repeating does not help at all.

Reduce it by checking zero before you start, calibrating, and using a set square or eye-level reading to avoid parallax. Using a graph's **gradient** also cancels a constant offset.

Reading instruments

You need to know the **resolution** of the instruments you'd choose. A metre rule reads to 1 mm, **vernier calipers** to 0.1 mm and a **micrometer screw gauge** to 0.01 mm. In the exam you may be shown a picture of one and asked for the reading to the correct number of significant figures.

Instrument	Resolution	Typical use
Metre rule	1 mm	Lengths of a pendulum or wire, heights, extensions over 1 cm
Vernier calipers	0.1 mm	Diameter of a ball bearing or cylinder, internal diameters, depths
Micrometer screw gauge	0.01 mm	Diameter of a wire (CP3, CP7), thickness of a sheet
Stopwatch	0.01 s (but human reaction time ≈ 0.1–0.3 s)	Timing 10 or 20 oscillations

Light gate + data logger	~0.001 s	Short times and speeds (CP1, CP9)
Digital meters	the last digit shown	Current, p.d., mass (top-pan balance)

Vernier calipers

The sliding vernier scale has 10 divisions that span 9 mm, so each one is 0.9 mm long. Every vernier line is 0.1 mm shorter than a main-scale millimetre, so when the jaws open by an extra 0.1 mm, the next vernier line lines up with a main-scale line. To read it:

1. Read the main scale at the vernier's **zero**: whole millimetres.
2. Find the vernier line that lines up **exactly** with a main-scale line. Its number is the tenths of a millimetre.
3. Add them, and check the zero reading with the jaws closed (zero error).

Micrometer screw gauge

One full turn of the thimble moves the spindle 0.50 mm, and the thimble has 50 divisions, so each division is 0.01 mm. Read the **sleeve** (millimetres and the half-millimetre marks below the line) and add the **thimble** division on the datum line × 0.01 mm. Always use the ratchet so you don't squash the wire, and check the zero.

EXAM TIP

When measuring a wire's diameter, take readings at **several places along the wire** and at **right angles** to each other (in case it isn't perfectly round), then average. Write it in the method; it's an easy mark in almost every planning question.

Uncertainties

Uncertainty in a single reading or in repeats

- **Single reading (or all repeats the same):** uncertainty = **half the resolution**. A ruler marked in mm reads to ± 0.5 mm. A digital ammeter showing 0.4 A (resolution 0.1 A) is ± 0.05 A.
- **Repeated readings:** uncertainty = **half the range** of the repeats. Edexcel also accepts the difference between the mean and the reading furthest from it.
- **Systematic offset** (e.g. a known zero error): don't add it to the uncertainty, **subtract it from each reading**.

Percentage uncertainty

$$\text{percentage uncertainty} = \frac{\text{uncertainty}}{\text{measured value}} \times 100\%$$

Usually quoted to 1 or 2 significant figures
Below about 5% suggests the measurement is repeatable

WORKED EXAMPLE

The diameter of a canister is measured three times with a ruler: 66 mm, 64 mm and 61 mm. Give the diameter with its uncertainty and its percentage uncertainty.

Show solution

Mean = (66 + 64 + 61)/3 = 63.7 mm. To the resolution of the ruler that's 64 mm.

$$\text{uncertainty} = \tfrac{1}{2}(\text{range}) = \tfrac{1}{2}(66 - 61) = 2.5 \text{ mm}$$

$$\text{percentage uncertainty} = \frac{2.5}{64} \times 100\% = 3.9\% \approx 4\%$$

So the diameter is 64 mm ± 2.5 mm (± 4%). The **radius** has the same percentage uncertainty, 4%, even though its absolute uncertainty is halved to 1.25 mm.

Combining uncertainties UNIT 6

When you calculate something from several measurements, their uncertainties combine. There are just three rules:

Adding or subtracting

Add the **absolute** uncertainties.

$$(102.8 \pm 0.3) - (101.4 \pm 0.1) = 1.4 \pm 0.4 \text{ mm}$$

Multiplying or dividing

Add the **percentage** uncertainties.

$$\%\,\rho = \%\,m + \%\,V$$

Raising to a power

Multiply the percentage uncertainty **by the power**.

$$\%(d^2) = 2 \times \% \, d$$

COMMON MISTAKE

Subtracting two nearly equal measurements is dangerous: the **absolute** uncertainties still add, but the result is small, so its **percentage** uncertainty is huge. In the pipe example above, 0.4 mm on 1.4 mm is a 29% uncertainty from two readings that were each better than 0.3%.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Why percentage uncertainties add when you multiply

You only need to *use* the rules, but here's where they come from.

The multiply/divide rule NOT PART OF THE EDEXCEL SPEC

1. Let $Q = AB$, with A uncertain by ΔA and B by ΔB. The largest value Q could take is:

$$Q + \Delta Q = (A + \Delta A)(B + \Delta B) = AB + A\,\Delta B + B\,\Delta A + \Delta A\,\Delta B$$

2. Subtract $Q = AB$. The last term is the product of two small uncertainties, so it's tiny and can be ignored:

$$\Delta Q \approx A\,\Delta B + B\,\Delta A$$

3. Divide both sides by $Q = AB$:

$$\frac{\Delta Q}{Q} \approx \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

Fractional (and so percentage) uncertainties add. For $Q = A^n$ you're multiplying n copies of A, which is why the percentage is multiplied by n.

4. Statisticians go one step further: independent random errors partly cancel, so they combine them "in quadrature", $\frac{\Delta Q}{Q} = \sqrt{(\frac{\Delta A}{A})^2 + (\frac{\Delta B}{B})^2}$. Edexcel uses the simple sum, which is a safe upper estimate.

Recording results and significant figures

- Column headings carry the **quantity and unit**, separated by a solidus: *length / m*, *time for 10 oscillations / s*. Never put units in the body of the table.
- Raw readings go to the **resolution of the instrument**, with the same number of decimal places all the way down a column (write 0.50 m, not 0.5 m, if the rule reads to 0.01 m).
- Processed values (like T^2 or $1/V$) usually go to **3 significant figures**, about what you can plot on a graph. A final answer should have the same number of s.f. as the **least precise** data used, normally 2 or 3.
- Take enough readings: usually at least **6 different values** over as **wide a range** as possible, with repeats, and look for any anomalous result to repeat.

QUICK CHECK

A student times 10 oscillations of a pendulum four times: 15.2 s, 15.6 s, 15.3 s and 15.5 s. What is the period, with its uncertainty?

- 1.54 s ± 0.4 s
- 15.4 s ± 0.2 s
- 1.54 s ± 0.02 s
- 1.540 s ± 0.005 s

Mean of 10T = 15.4 s with half-range uncertainty (15.6 − 15.2)/2 = 0.2 s. Dividing by 10 gives T = 1.54 s ± 0.02 s. Timing 10 oscillations divides the reaction-time uncertainty by 10, which is exactly why we do it.

Straight-line graphs

Almost every experiment ends with a straight-line graph, because a straight line is easy to recognise, a line of best fit averages out random error, and its **gradient** and **intercept** give you the constant you're after. The trick is to rearrange the physics equation into the form

$$y = m\,x + c$$

and then plot whatever plays the role of y against whatever plays the role of x.

Equation	Plot y	against x	Gradient	Intercept
$h = \frac{1}{2}gt^2$ (CP1)	h	t^2	$g/2$	0

$R = \rho l/A$ (CP7)	R	l	ρ/A	0 (contact resistance if not)
$V = \varepsilon - Ir$ (CP8)	V	I	$-r$	ε
$T = 2\pi\sqrt{m/k}$ (CP16)	T^2	m	$4\pi^2/k$	0
$pV = NkT$ at fixed T (CP14)	p	$1/V$	NkT	0
$V = V_0 e^{-t/RC}$ (CP11)	$\ln V$	t	$-1/RC$	$\ln V_0$

- Choose scales so the points fill **more than half** the grid in both directions, with sensible steps (1, 2 or 5 per square; never 3).
- Plot points as small crosses to within half a small square. Draw **one** thin straight line of best fit with roughly equal numbers of points either side. Don't force it through the origin, and never join the dots.
- Find the gradient from a **large triangle** (at least half the length of the line) using points **on your line**, not data points. Show the triangle on the graph.
- A line that should pass through the origin but doesn't suggests a **systematic error**. A large scatter of points suggests large **random errors**.

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Error bars and worst-fit lines

Edexcel says there is **no expectation** that error bars are added to graphs. In university labs (and some other exam boards) you draw an error bar for each point, then the steepest and shallowest lines that still pass through all the bars. Half the difference between those gradients is the uncertainty in the gradient. It's a good way to see how much a graph can really tell you, but you won't be asked to do it.

Logarithmic graphs UNIT 6

Two kinds of relationship can't be straightened just by squaring or inverting, because you don't know the power yet, or the variable sits in an exponent. Taking logs turns both into straight lines.

Straightening a power law $y = kx^n$

1. Take logs (base 10 or base e, as long as you are consistent) of both sides:

$$\log y = \log(kx^n)$$

2. Use $\log(ab) = \log a + \log b$ and $\log(x^n) = n \log x$:

$$\log y = n \log x + \log k$$

3. Compare with $y = mx + c$: plotting $\log y$ against $\log x$ gives a straight line with

$$\text{gradient} = n, \qquad \text{intercept} = \log k$$

So a log–log graph tells you the **power** directly. For example, plotting $\log T$ against $\log l$ for a pendulum gives a gradient of 0.5.

Straightening an exponential y = A e^(−kx)

1. Take natural logs (base e) of both sides:

$$\ln y = \ln\left(Ae^{-kx}\right) = \ln A + \ln\left(e^{-kx}\right)$$

2. Since $\ln(e^z) = z$:

$$\ln y = -k\,x + \ln A$$

3. Plot $\ln y$ against x (only y is logged): a straight line with gradient $-k$ and intercept $\ln A$. This is how you get the time constant from capacitor discharge (CP11) and the decay constant in radioactivity.

EXAM TIP

Logs have no units, so label the axis *ln (V / V)* or *log (T / s)*: take the log of the number, not the unit. When reading a gradient on a log graph, use differences in the **log values** on the axes, not the original numbers.

Drawing conclusions

To decide whether your result is **accurate**, compare it with the accepted value in one of two ways:

If you know the percentage uncertainty

Work out the range your result could lie in. If the accepted value is inside it, the result is accurate. Example: $g = 10.1 \text{ m s}^{-2} \pm 4\%$ means 9.7 to 10.5 m s⁻², which includes 9.81, so it's accurate.

If you don't (e.g. a value from a gradient)

Find the percentage difference. Less than 5% indicates the result is accurate.

$$\frac{10.1 - 9.81}{9.81} \times 100\% = 3\%$$

Then go further than "it was accurate". Say which measurement had the largest percentage uncertainty, and suggest a **specific** improvement: "use a micrometer instead of a ruler for the diameter", not "use better equipment". Mention any systematic error suggested by a non-zero intercept.

Planning an experiment

Planning questions are marked against a checklist. Hit every point:

1. **Variables:** state the independent variable (what you change), the dependent variable (what you measure) and every control variable, with *how* each is kept constant.
2. **Apparatus with resolution:** name each instrument and its resolution, chosen to suit the size of the measurement (micrometer for a wire, metre rule for its length).
3. **Method:** how each variable is measured, including technique details (eye level to avoid parallax, set square, fiducial marker at the centre of an oscillation, timing 20 oscillations, checking zero).
4. **Range and repeats:** at least 6 values over a wide range, repeated and averaged where readings vary.
5. **Analysis:** which graph you'll plot and how the gradient or intercept gives the answer.
6. **Sources of uncertainty** and how to reduce them, plus any **safety** issue with a sensible precaution (goggles when wires may snap, a sand tray under hanging masses, lead-lined handling tongs for sources).

The 16 core practicals

Each one is covered on its topic page with a virtual version you can run. Click through to revise the method and analysis.

#	Core practical	Key graph or analysis
1	Acceleration of a freely-falling object	h against t^2, gradient $= g/2$
2	Viscosity of a liquid by the falling-ball method	Terminal velocity from time between marks, then Stokes' law
3	Young modulus of a material	Stress against strain, gradient $= E$
4	Speed of sound with a 2-beam oscilloscope	Move the microphone through whole wavelengths, then $v = f\lambda$
5	Frequency of a vibrating string: length, tension, mass per unit length	f against $1/l$, or f against $\sqrt{T}$
6	Wavelength of light with a diffraction grating	$n\lambda = d\sin\theta$ for several orders

7	Resistivity of a material	R against l, gradient $= \rho/A$
8	E.m.f. and internal resistance of a cell	V against I: intercept ε, gradient $-r$
9	Force and change of momentum	Impulse $F\Delta t$ compared with Δp
10	Collisions between small spheres using ICT	Momentum components before and after
11	Capacitor charge and discharge on an oscilloscope or data logger	$\ln V$ against t, gradient $= -1/RC$
12	Calibrate a thermistor in a potential divider as a thermostat	Calibration curve of V_{out} against temperature
13	Specific latent heat of a phase change	$L = \Delta E/\Delta m$
14	Pressure and volume of a gas at fixed temperature	p against $1/V$, a straight line through the origin
15	Absorption of gamma radiation by lead	Corrected count rate against thickness
16	Unknown mass from resonant frequencies of known masses	T^2 (or $1/f^2$) against m

Quick checks

QUICK CHECK

A metre rule has a zero error: it reads 2 mm when it should read zero. What kind of error is this, and what should you do?

- Random error: take more repeats and average.
- Systematic error: subtract 2 mm from every reading.
- Systematic error: add 2 mm to the uncertainty.
- Random error: it cancels when you average.

A zero error shifts every reading by the same amount in the same direction, so it is systematic. You correct each reading. Repeating can't remove it.

QUICK CHECK

A wire's diameter is measured as 0.50 mm ± 0.01 mm. What is the percentage uncertainty in its cross-sectional area?

- 1%
- 2% × π
- 4%
- 0.02%

% uncertainty in d = 0.01/0.50 × 100% = 2%. Area $= \pi d^2/4$, so the percentage doubles to 4%. π and the 4 are exact numbers with no uncertainty.

QUICK CHECK

The intensity of a source falls with distance as $I = kd^n$. Which graph gives n as its gradient?

- I against d
- $\ln I$ against d
- $\log I$ against $\log d$
- I against $1/d^2$

$\log I = n \log d + \log k$, a straight line with gradient n. Plotting I against $1/d^2$ only works if you already know that $n = -2$.

QUICK CHECK

A student repeats a measurement many times and gets very similar values, but they are all well away from the accepted value. The results are best described as:

- accurate and precise
- precise but not accurate
- accurate but not precise
- valid and reproducible

Close agreement between repeats means precise. Being far from the true value means not accurate, which points to a systematic error.